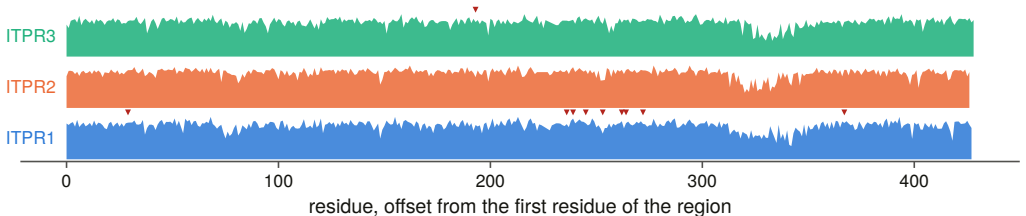
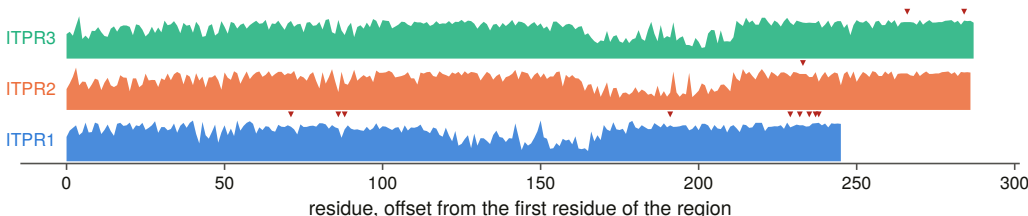


a the IP3-binding core (β -trefoil + MIR) carries deep-layer constraint; each marker is a pathogenic position



b the pore module spans channel, filter, gate and luminal loop



c every pathogenic position in the two regions is listed with the residue each paralogue carries there

		ITPR1/2/3						ITPR1/2/3			
		1	2	3				1	2	3	
ITPR1 34	β-trefoil (PF08709)	D	D	D	→N	ITPR1 2449	channel (TM)	I	I	I	→V
ITPR3 196	β-trefoil (PF08709)	A	A	A	→T	ITPR1 2451	channel (TM)	L	L	L	→P
ITPR1 241	MIR	R	R	R	→K	ITPR1 2592	channel (TM)	L	L	L	→P
ITPR1 244	MIR	H	H	H	→P	ITPR1 2600	channel (TM)	T	T	T	→P
ITPR1 250	MIR	F	F	F	→L	ITPR1 2601	channel (TM)	F	F	F	→L
ITPR1 258	MIR	K	K	G	→M	ITPR3 2506	channel (TM)	I	I	I	→N
ITPR1 267	MIR	T	T	T	→R	ITPR3 2524	channel (TM)	R	R	R	→H
ITPR1 269	MIR	R	R	R	→L	ITPR1 2554	filter	G	G	G	→R
ITPR1 277	MIR	S	S	S	→I	ITPR2 2498	filter	G	G	G	→S
ITPR1 372	MIR	P	A	P	→L	ITPR1 2595	gate	G	G	G	→A
ITPR1 2434	channel (TM)	T	T	T	→I	ITPR1 2598	gate	I	I	I	→T

a red letter marks a residue the other paralogue does not share; every letter was checked against that paralogue's own per-residue table